

Customer Behavior Analysis

Segmentation, Purchase Patterns & Churn Risk — Prepared for Alfido Tech

49,661

Customers

250,000

Transactions

\$681,346,299

Total Revenue

\$2,725

Avg. Order Value

Data period: 2020-01-01 to 2023-09-13 | Source: Kaggle — Customer Behavior Analysis dataset
Prepared as part of a Data Analytics Internship Project
[Your Name] | [Date]

1. Executive Summary

This report analyzes 250,000 e-commerce transactions from 49,661 customers (2020-01-01 to 2023-09-13) to segment customers, understand purchase patterns, and assess churn risk for Alfido Tech. Using RFM (Recency, Frequency, Monetary) analysis validated with K-Means clustering, customers were grouped into seven actionable segments. **Champions and Loyal Customers make up 38% of the customer base but generate 56% of total revenue**, while a substantial "Hibernating / Lost" segment (19% of customers) has not purchased in over a year and a half on average. A supervised churn model found the dataset's provided Churn label carries no predictive signal from purchase behavior — a transparent, evidence-based finding detailed in Section 5 — so we recommend Alfido Tech adopt a recency-based engagement-risk framework instead. Five actionable recommendations are provided in Section 7.

2. Methodology

Data cleaning: removed a duplicate age column, parsed transaction dates, treated missing return flags as "not returned," and validated there were no duplicate transactions or non-positive price/quantity/amount values (none were found — the dataset was clean going in).

Feature engineering: transactions were aggregated to one row per customer, computing Recency (days since last purchase), Frequency (number of transactions), Monetary value (total spend), average order value, return rate, tenure, and preferred category/payment method.

Segmentation: each customer was scored 1-5 on R, F, and M (quintiles) and mapped to one of seven business segments using standard RFM heuristics (Champions, Loyal Customers, Potential Loyalists, New Customers, Need Attention, At Risk, Hibernating/Lost). This was cross-validated with K-Means clustering on standardized, log-transformed RFM features.

Tools used: Python (pandas, scikit-learn, matplotlib/seaborn) for cleaning, feature engineering, clustering, and visualization; SQL (SQLite) for a parallel analytical query layer; Power BI-ready data exports for interactive dashboarding.

3. Customer Segments (RFM)

Segment	Customers	% of Base	Avg. Recency (days)	Avg. Frequency	Avg. Monetary	Total Revenue	% Revenue	Churn Rate*
Champions	8,808	17.7%	61	7.6	\$21,656	\$190,749,317	28.0%	21.0%
Loyal Customers	10,157	20.5%	260	6.9	\$18,693	\$189,868,835	27.9%	20.0%
Potential Loyalists	7,993	16.1%	128	4.6	\$12,492	\$99,845,294	14.7%	20.0%
Hibernating / Lost	9,369	18.9%	582	2.6	\$6,513	\$61,019,918	9.0%	21.0%
At Risk	4,098	8.3%	441	4.9	\$12,824	\$52,553,821	7.7%	19.0%
New Customers	5,400	10.9%	67	3.2	\$8,860	\$47,841,340	7.0%	19.0%
Need Attention	3,836	7.7%	315	3.6	\$10,289	\$39,467,774	5.8%	19.0%

*Churn Rate uses the dataset's provided label; see Section 5 for why this label should be treated with caution.

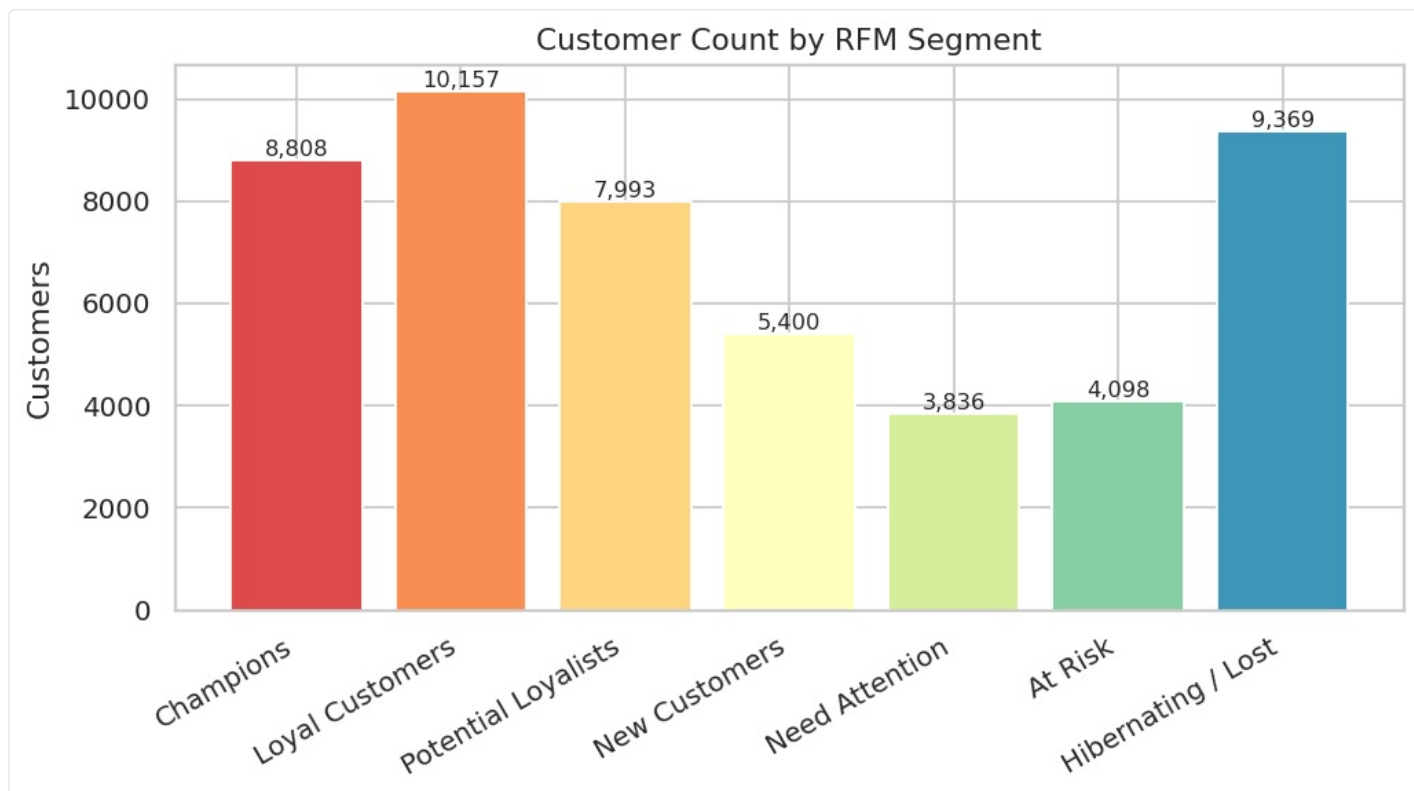


Figure 1. Customer count by RFM segment.

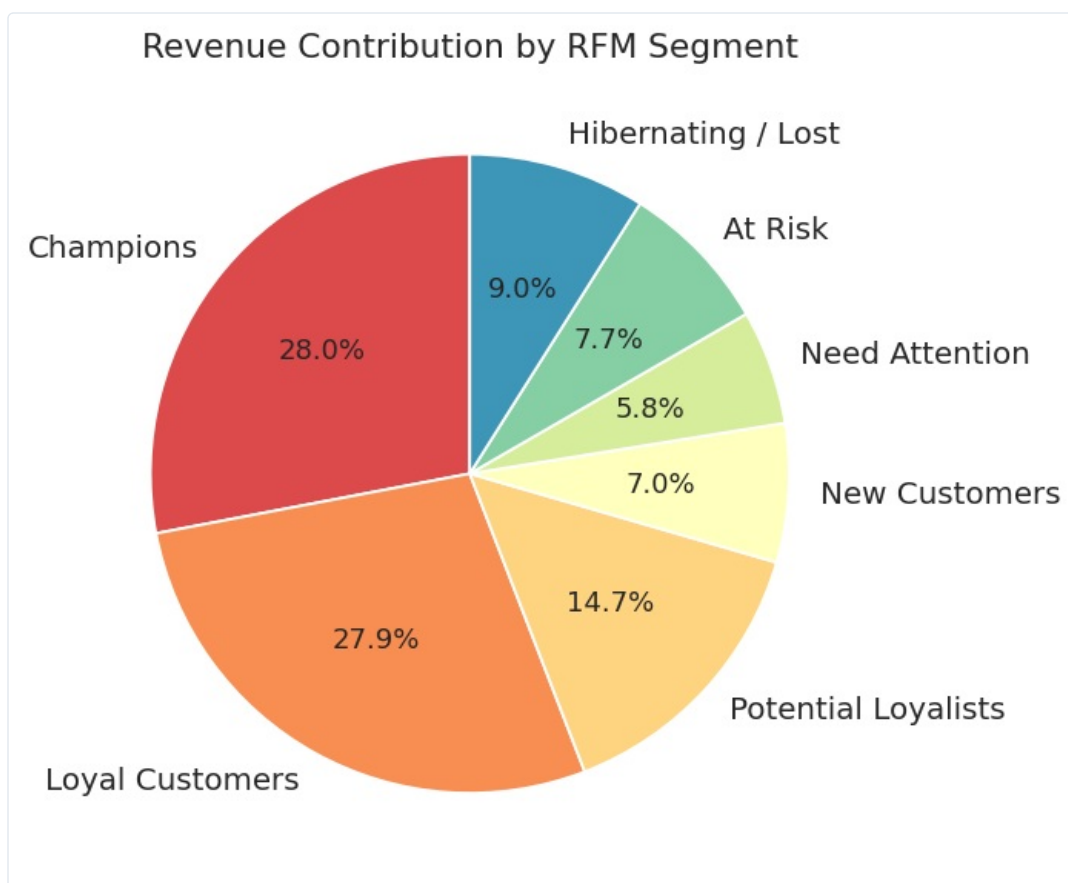


Figure 2. Revenue contribution by RFM segment — Champions and Loyal Customers drive the majority of revenue.

K-Means Clustering (validation)

An independent K-Means clustering (k=4, chosen for business interpretability; the silhouette score statistically favored k=2 — see notebook Section 5) produced groupings consistent with the RFM segmentation, cross-validating the approach:

Cluster	Customers	Avg. Recency	Avg. Frequency	Avg. Monetary	Churn Rate*

Loyal / Regular	8,181	554	4.3	\$12,054	20.0%
Growing	14,945	158	3.6	\$9,267	20.0%
At Risk	4,473	674	1.8	\$4,060	20.0%
Best Customers	22,062	141	6.9	\$19,313	20.0%

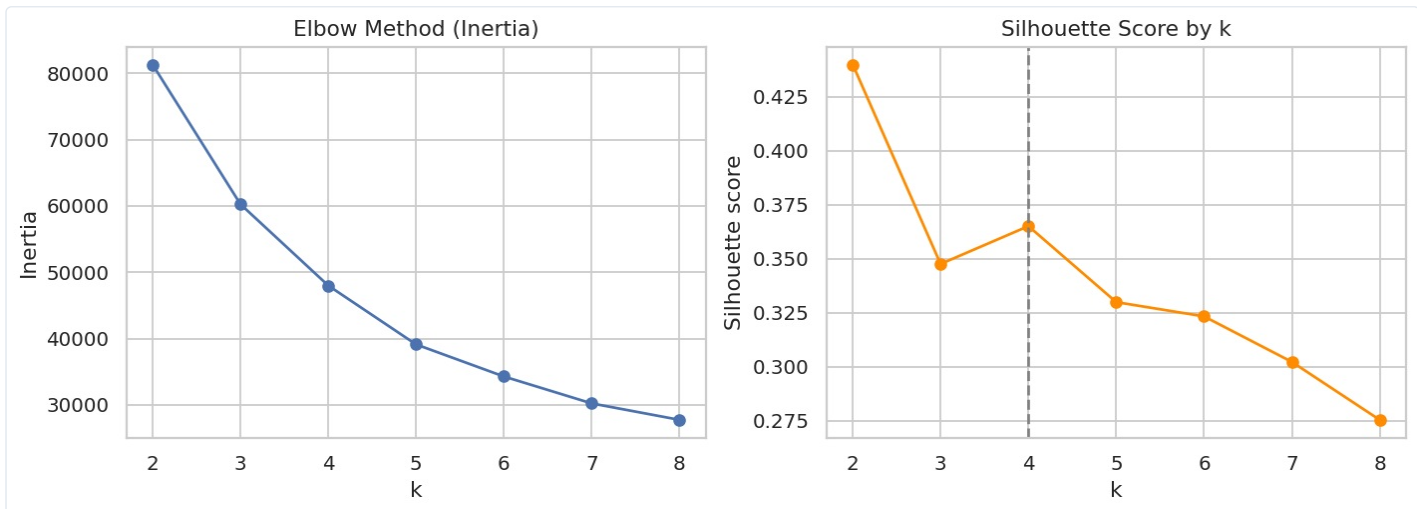


Figure 3. Elbow and silhouette diagnostics used to select the number of clusters.

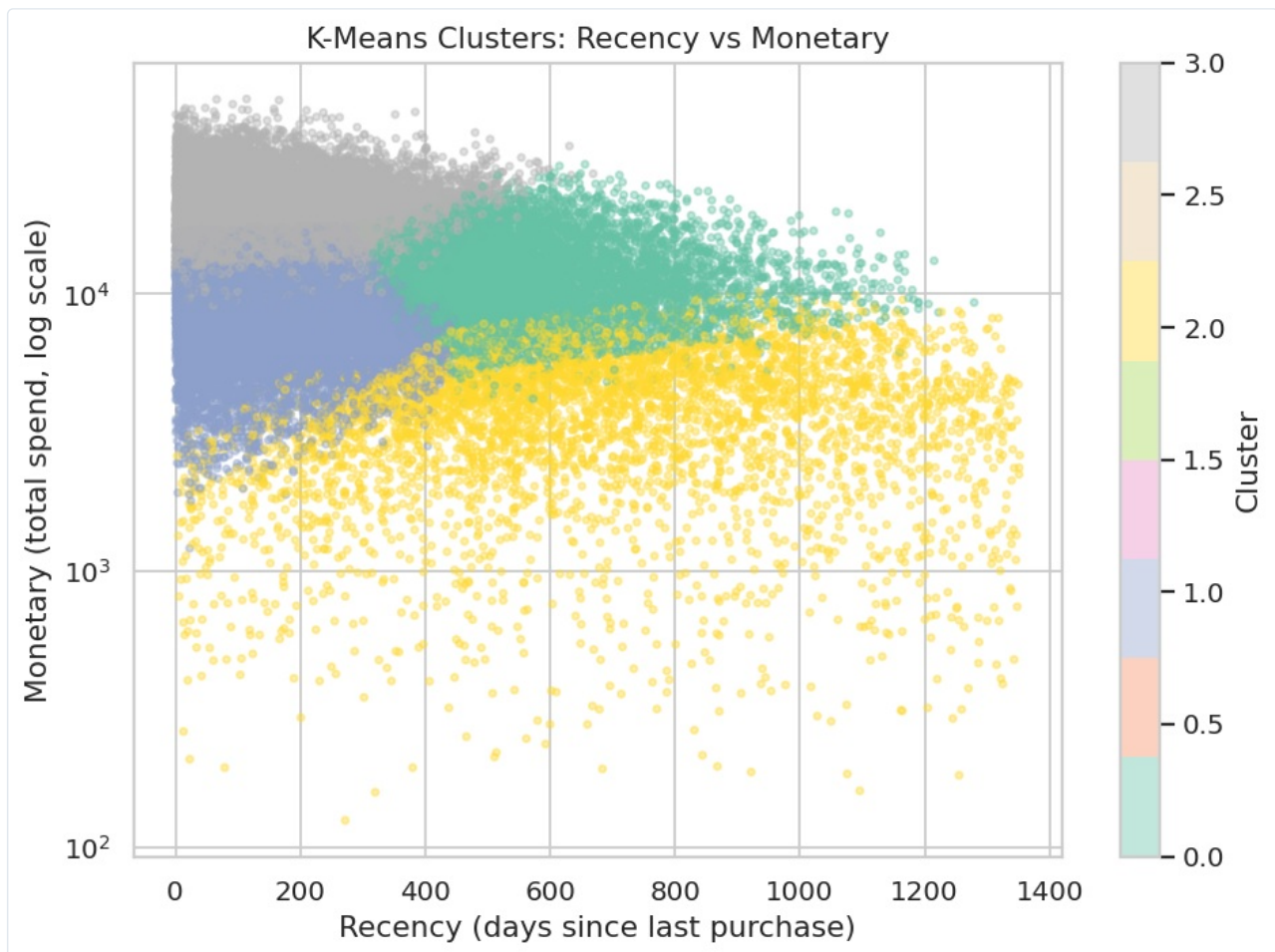


Figure 4. K-Means clusters plotted on Recency vs. Monetary value.

4. Purchase Patterns

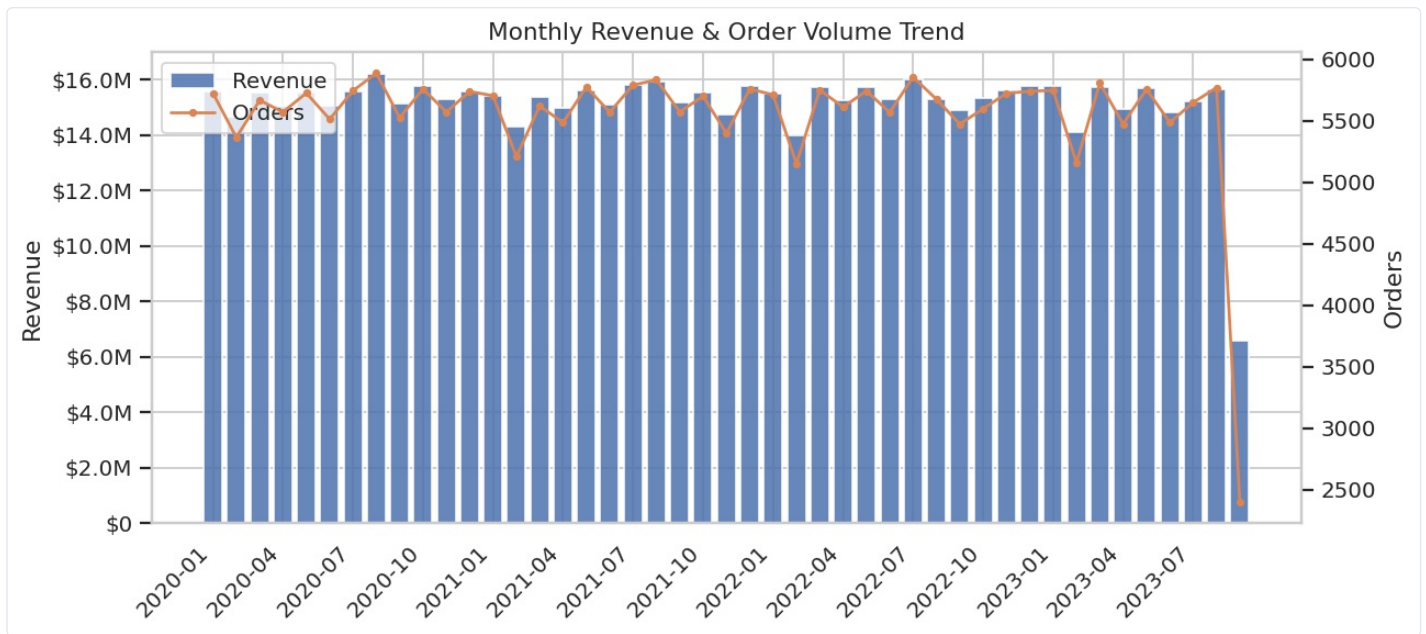


Figure 5. Monthly revenue and order volume, Jan 2020 – Sep 2023.

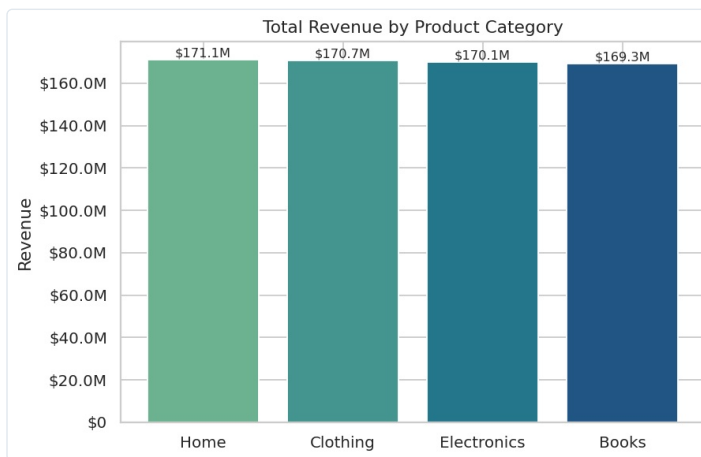


Figure 6. Revenue by product category.

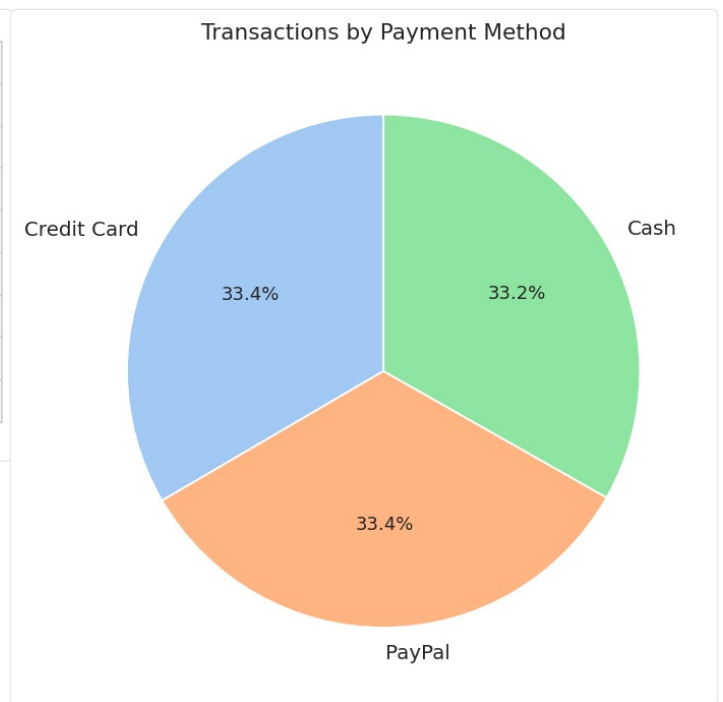


Figure 7. Transaction share by payment method.

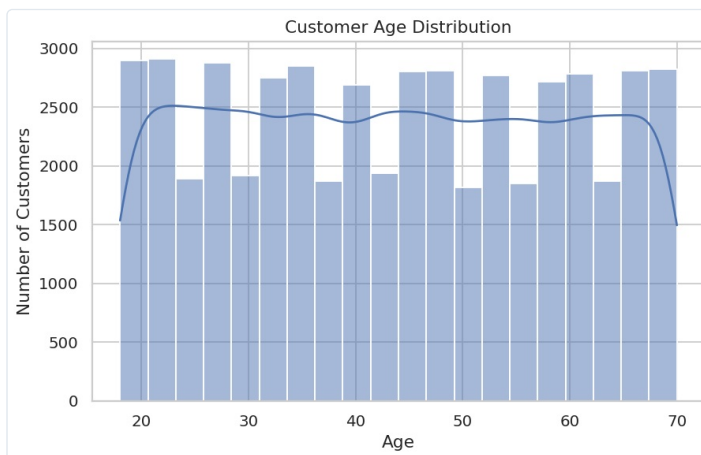


Figure 8. Customer age distribution.

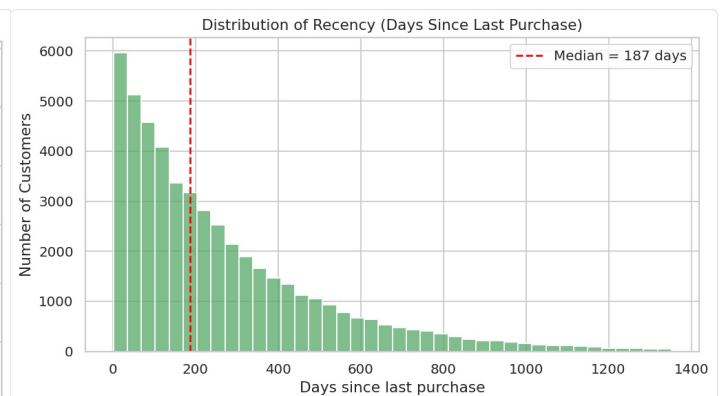


Figure 9. Distribution of recency across all customers.

5. Retention & Churn Risk

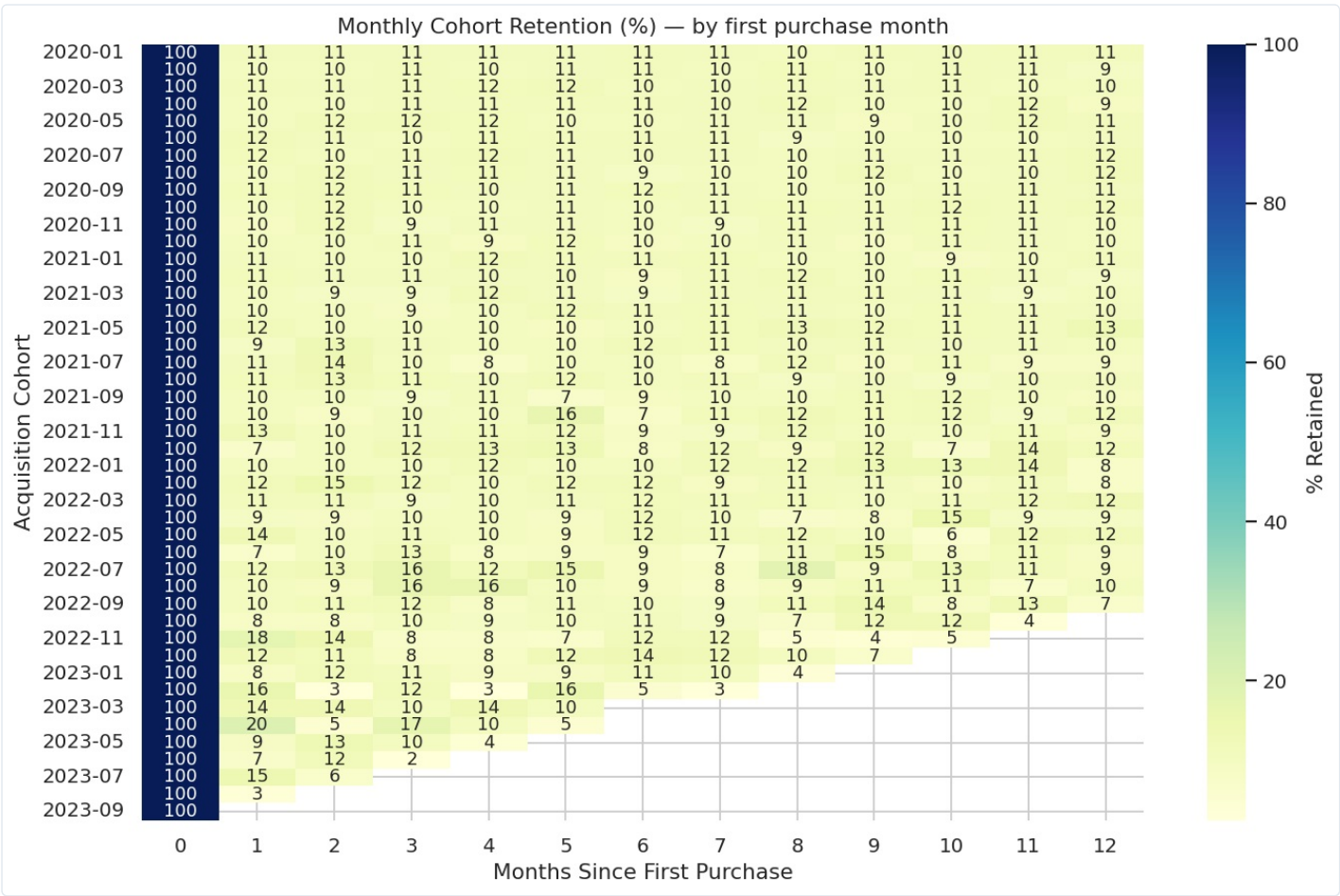


Figure 10. Monthly cohort retention heatmap.

Data quality finding: Retention flattens to roughly the same ~10-13% almost immediately after the acquisition month and stays flat through month 12 (Figure 10), rather than showing the gradual decay curve typical of real repeat-purchase behavior. We also tested the provided Churn label against every behavioral feature: correlations were all effectively zero ($|r| < 0.01$), and a logistic regression classifier scored **AUC ≈ 0.50** (no better than random guessing). This strongly suggests the churn label and long-run repurchase timing in this dataset are not behaviorally driven. We report this transparently rather than manufacturing a misleadingly confident churn model — the RFM segmentation itself remains valid and useful, since it is based on realized purchase behavior, not the churn label.

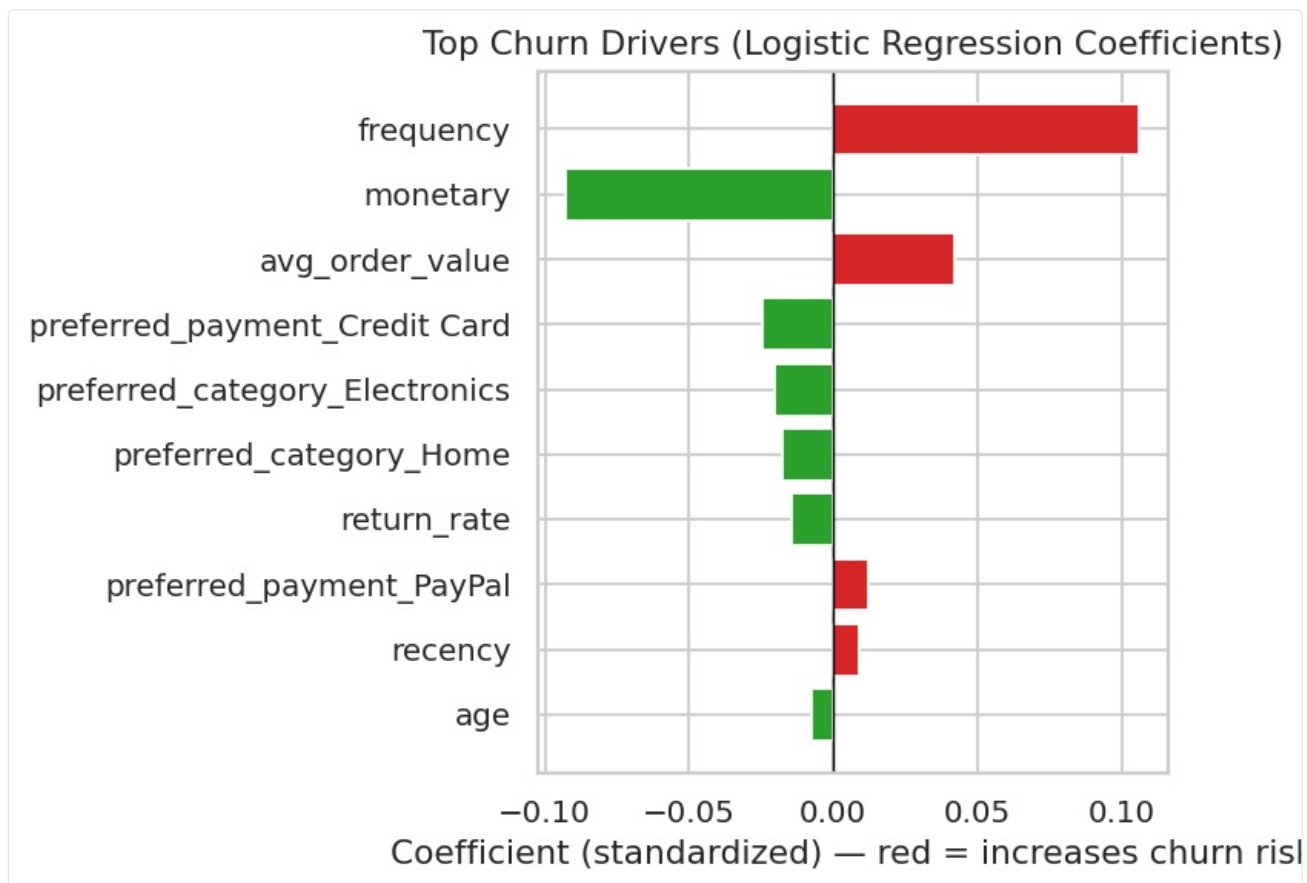


Figure 11. Logistic regression coefficients — even the largest are small and not statistically meaningful given the near-zero AUC.

Recommended operational churn-risk signal

Since the provided label isn't usable, we built a practical proxy: the share of each segment's customers who have gone quiet for 180+ days, prioritized by lifetime spend. This converts directly into a target list for retention campaigns (see the companion SQL script, Query 7, and the notebook Section 9 for the full ranked list).

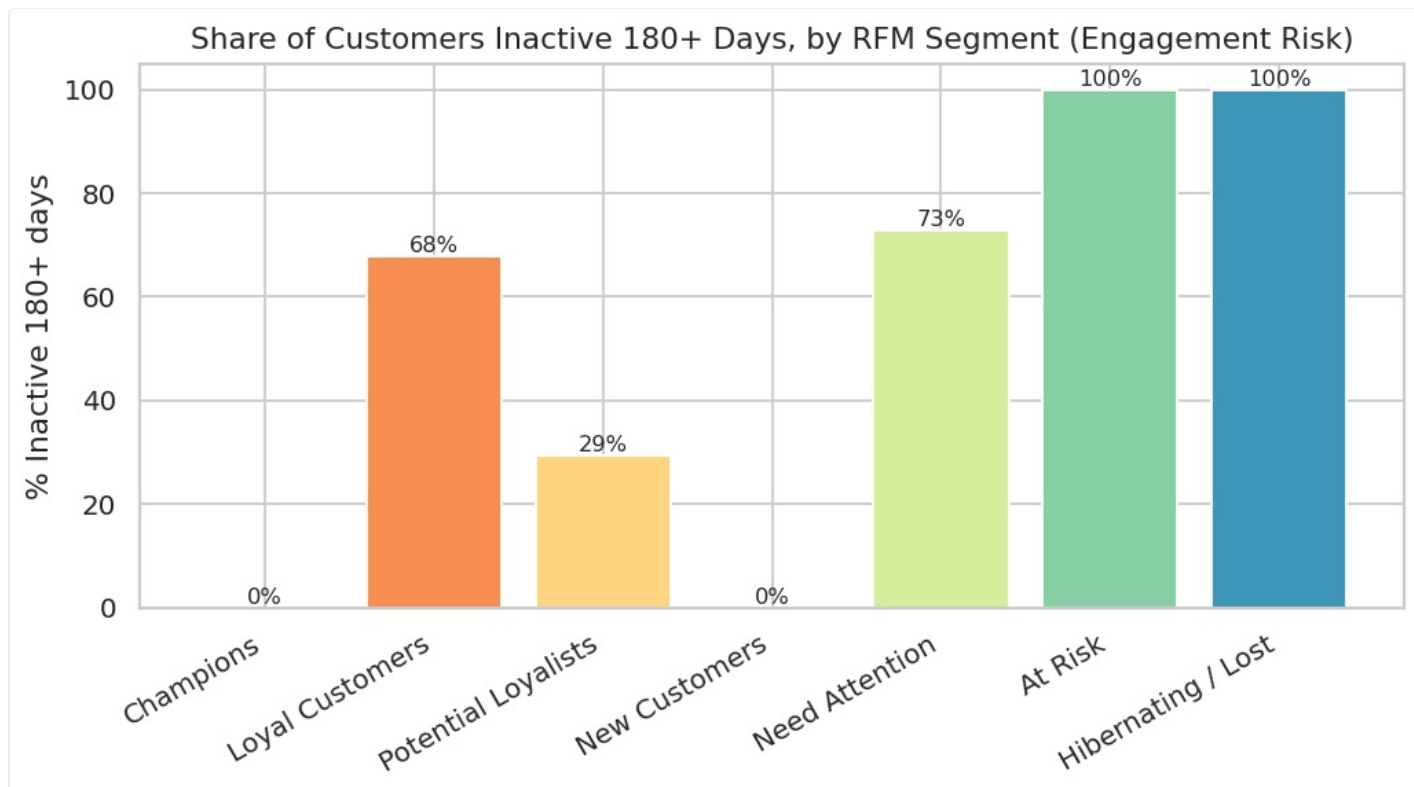


Figure 12. Share of customers inactive 180+ days, by RFM segment — this is the actionable 'at risk' view.

6. Deliverables Summary

Deliverable	File	What it contains
Jupyter Notebook	Alfido_Tech_Customer_Behavior_Analysis.ipynb	Full, executed code: cleaning, feature engineering, RFM, clustering, churn modeling, all charts
SQL Script	customer_behavior_analysis.sql	Schema + 10 analytical queries (segment summary, cohort retention, at-risk customers, etc.)
SQLite Database	alfido_customer_behavior.db	Cleaned transactions + customer feature tables, ready to query
Power BI Data Model	Alfido_PowerBI_Data_Model.xlsx	Star-schema-ready fact/dimension tables (Fact_Transactions, Dim_Customers, Dim_Date, Segment_Summary)
PDF Report	this document	Executive summary, findings, recommendations

See the companion **Power BI Build Guide** for the specific visuals and DAX measures to recreate this analysis as an interactive dashboard.

7. Five Actionable Recommendations for Alfido Tech

1. Launch a targeted win-back campaign for high-value at-risk customers

Prioritize the ranked list of top-spending customers inactive 180+ days (SQL Query 7 / notebook Section 9). Personalized outreach or a meaningful discount for this narrow, high-LTV group will recover more revenue per dollar spent than a blanket promotion.

2. Build a formal VIP / loyalty tier for Champions and Loyal Customers

This 38% of the customer base drives 56% of revenue. Early access to new products, free shipping, or a points program protects this base from competitor poaching and increases share-of-wallet.

3. Automate lifecycle marketing by RFM segment instead of blanket campaigns

Onboarding nudges for New Customers, replenishment reminders for Potential Loyalists, and reactivation offers for Hibernating customers will outperform one-size-fits-all email blasts.

4. Investigate the ~40% product return rate with the fulfillment/quality team

Return rates are unusually high and nearly identical across every category and payment method, suggesting a systemic issue (packaging, sizing/description accuracy, fulfillment errors) rather than a category-specific one. Even a modest reduction meaningfully improves net revenue and customer satisfaction.

5. Replace the current Churn field with a live, behavior-based risk score in the CRM

Since the existing label carries no predictive signal, adopt the recency/frequency/monetary risk framework built in this project as the operational "churn risk" metric going forward, and re-validate it periodically against actual cancellations/lapses once that ground truth exists.

8. Conclusion

Alfido Tech's customer base is highly concentrated: a relatively small group of engaged, frequent, high-spending customers accounts for the majority of revenue, while a large dormant segment represents a significant win-back opportunity. Because the dataset's built-in churn label is not behaviorally meaningful, the most reliable and immediately actionable lens is the RFM segmentation built here, paired with a simple recency-based risk score. Implementing segment-specific retention and loyalty programs, and investigating the elevated return rate, are the highest-leverage next steps.